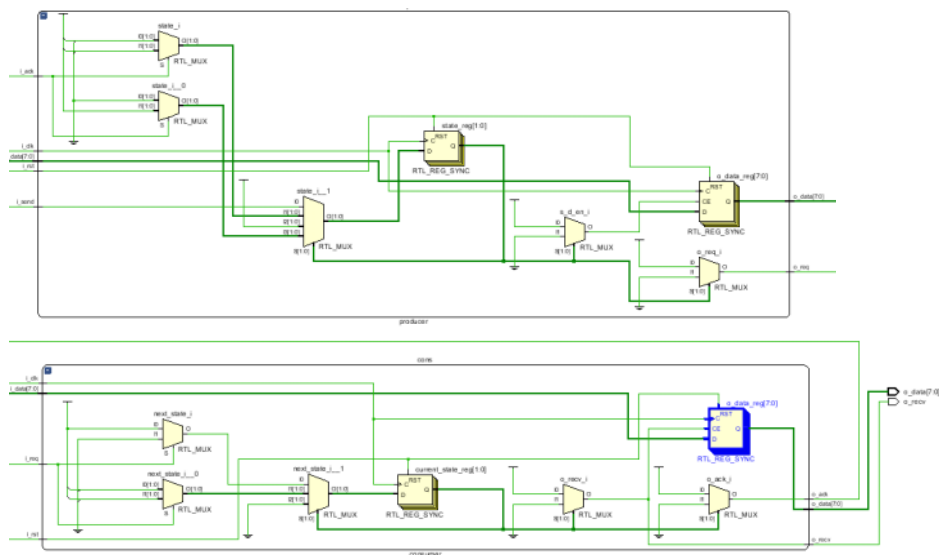
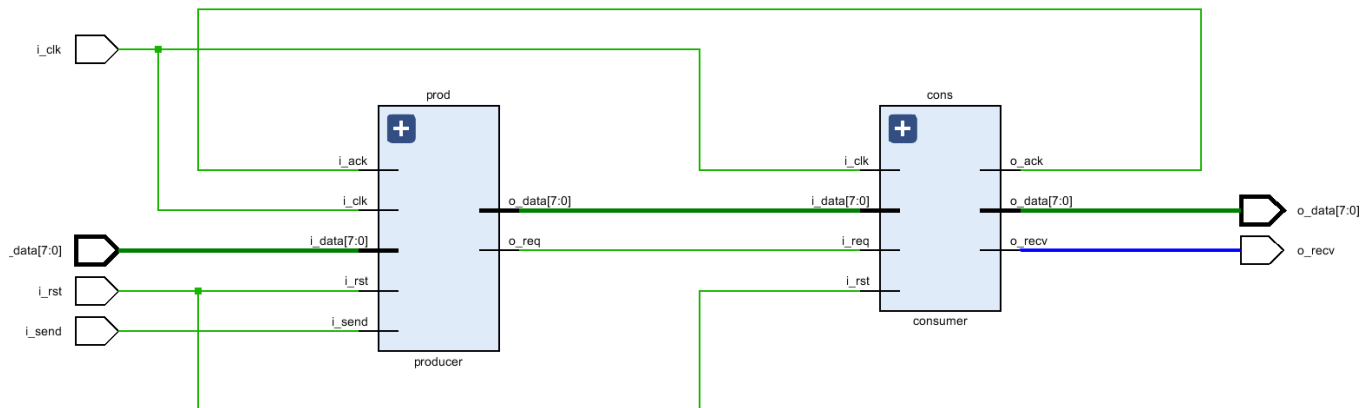


Tutorial 2

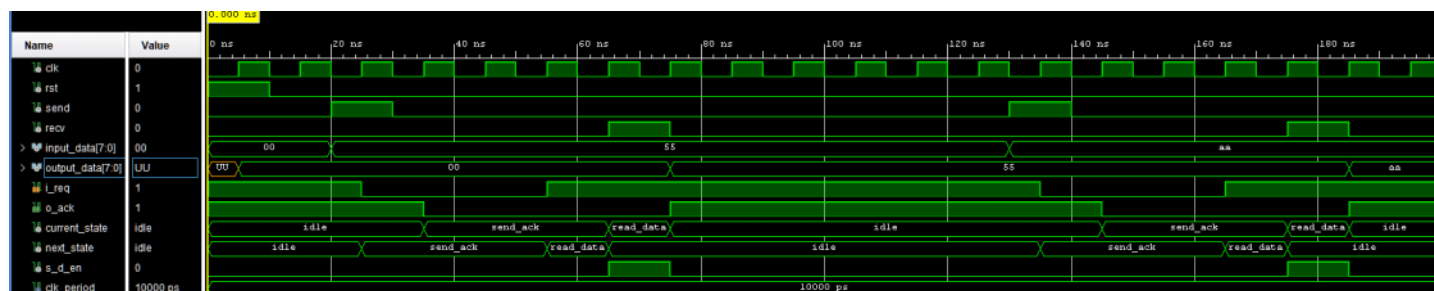
Friday, September 11, 2026 12:10 PM

Part1:



The first 2 muxes take in either req or ack and change the state. Then those states are fed into another mux that uses a regi ster with the current state as its select signal. That register also takes in the output of the last mux to replace the current state when the next rising edge of the clock cycle hits. The current state is also used for 2 other muxes that handle ack and rcv in consumer, and d_en and req for producer. Lastly, d_en and rcv, are used as inputs for the clocked register that stores o_data.

PART2:



At the start rst is set high to keep everything at idle. Then when send gets set high the input data is loaded into the bus ready for req and ack to be set low. First req is set low which sets consumer into send_ack, which sets ack low. That moves producer into send_data state, then into the hold_data which sets req high. This causes consumer to switch states to read_data and set rcv and d_en. This loads the data into the output bus. Then everything is repeated for the next data byte.

PART3:

INFO: [Synth 8-802] inferred FSM for state register 'state_reg' in module 'producer'
INFO: [Synth 8-802] inferred FSM for state register 'current_state_reg' in module 'consumer'

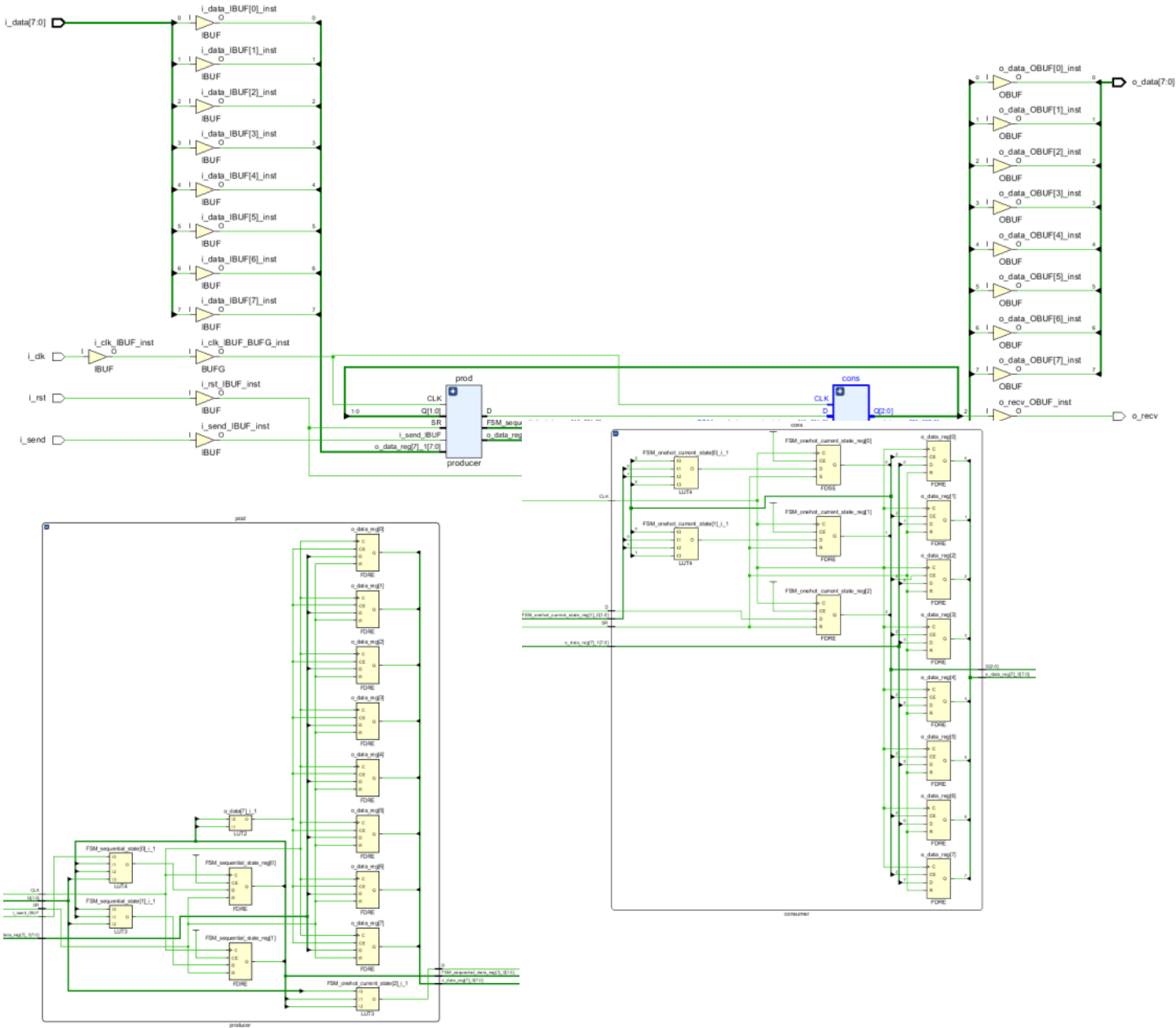
State	New Encoding	Previous Encoding
idle	00	00
ack_wait	01	01
send_data	10	10
hold_data	11	11

INFO: [Synth 8-3354] encoded FSM with state register 'state_reg' using encoding 'sequential' in module 'producer'

State	New Encoding	Previous Encoding
idle	001	00
send_ack	010	01
read_data	100	10

INFO: [Synth 8-3354] encoded FSM with state register 'current_state_reg' using encoding 'one-hot' in module 'consumer'

As can be seen Producer uses the sequential encoding while Consumer uses one-hot.



PART4:

One Hot:

INFO: [Synth 8-802] inferred FSM for state register 'state_reg' in module 'producer'
INFO: [Synth 8-802] inferred FSM for state register 'current_state_reg' in module 'consumer'

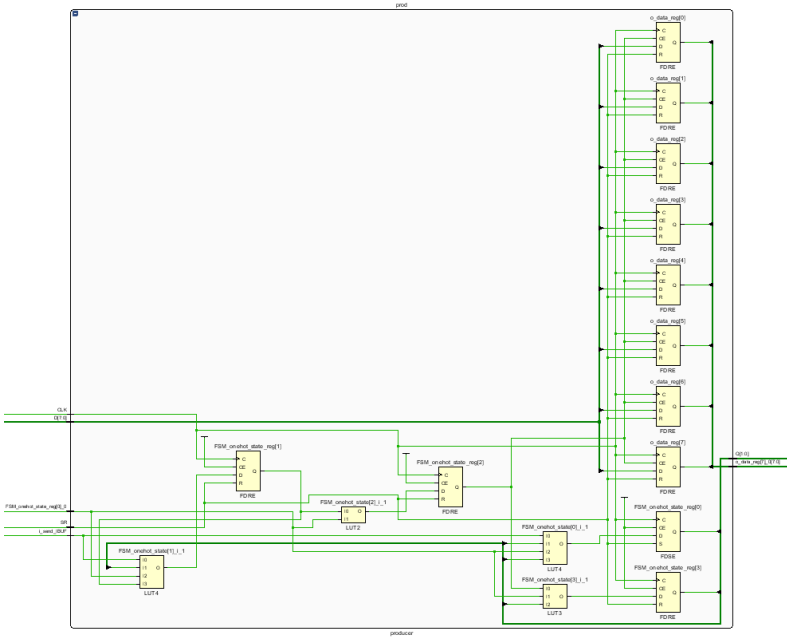
State	New Encoding	Previous Encoding
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idle	0001	00
ack_wait	0010	01
send_data	0100	10
hold_data	1000	11

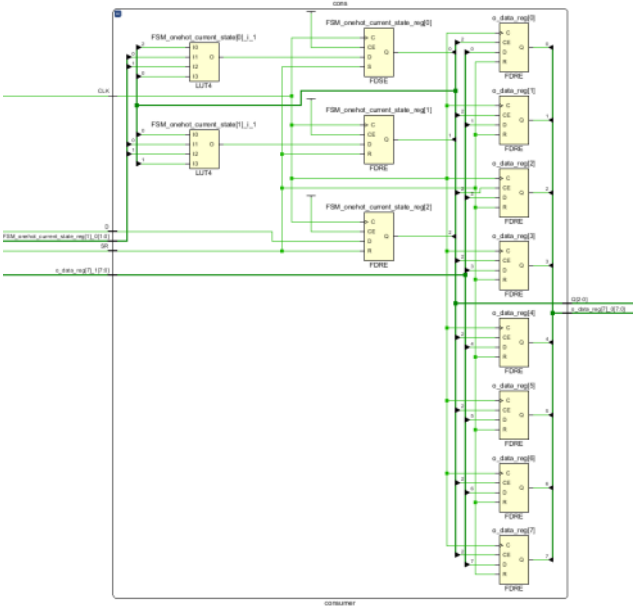
INFO: [Synth 8-3354] encoded FSM with state register 'state_reg' using encoding 'one-hot' in module 'producer'

State	New Encoding	Previous Encoding
idle	001	00
send_ack	010	01
read_data	100	10

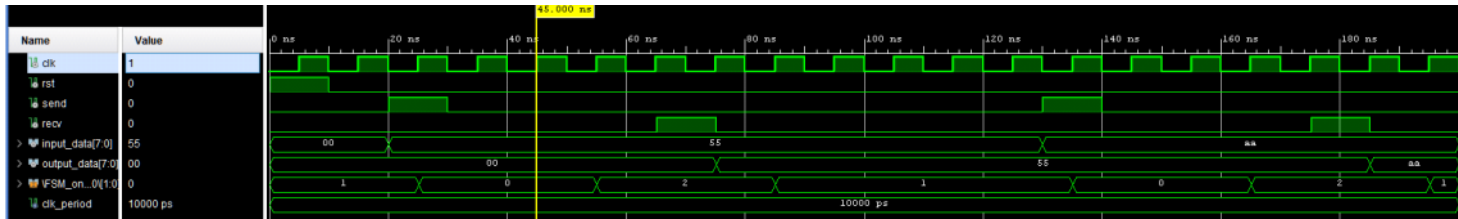
INFO: [Synth 8-3354] encoded FSM with state register 'current_state_reg' using encoding 'one-hot' in module 'consumer'



4 Luts, 8 data reg FFs, 6 state reg FFs



2 LUTs, 3 state reg FF, 8 data reg FF



Sequential:

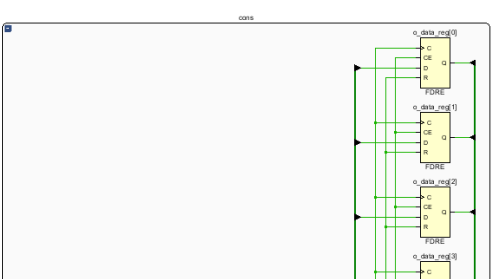
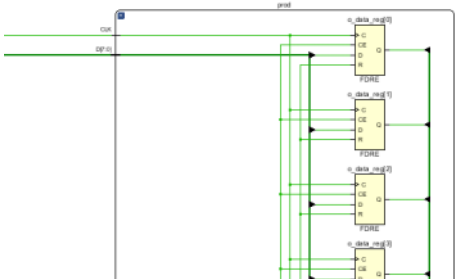
INFO: [Synth 8-802] inferred FSM for state register 'state_reg' in module 'producer'
 INFO: [Synth 8-802] inferred FSM for state register 'current_state_reg' in module 'consumer'

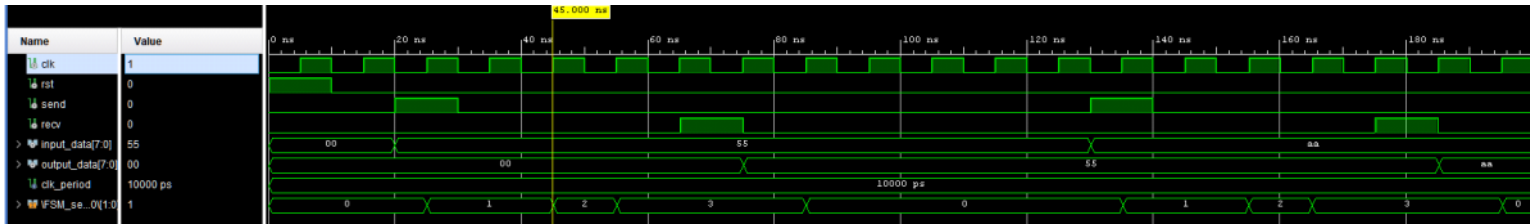
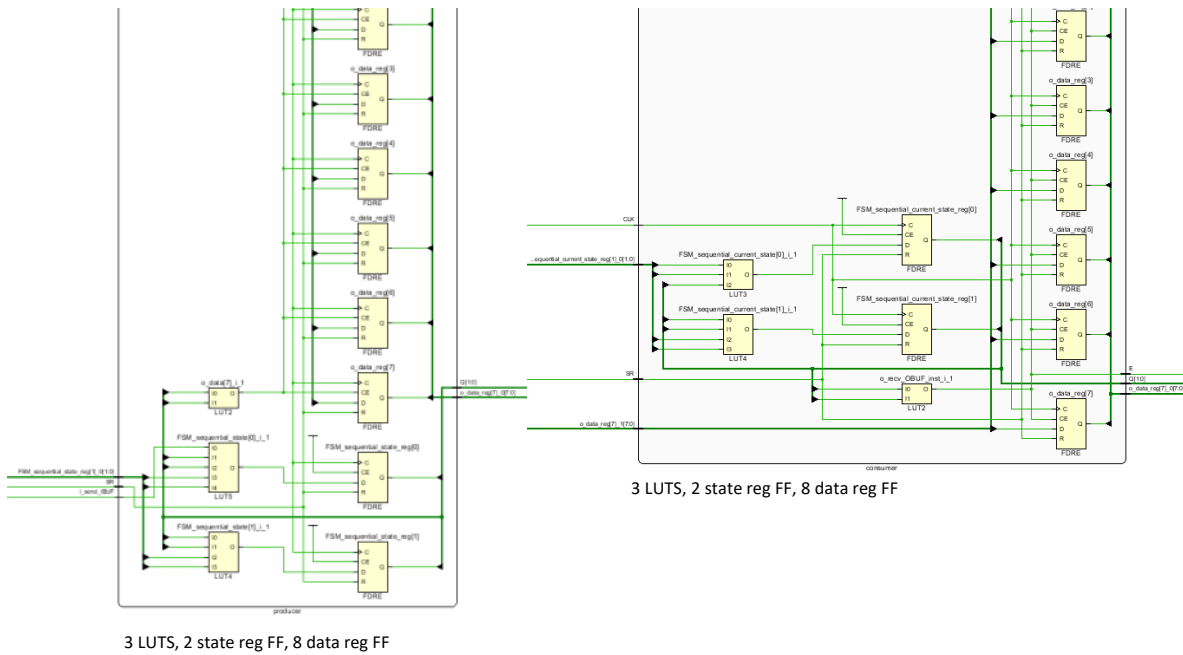
State	New Encoding	Previous Encoding
idle	00	00
ack_wait	01	01
send_data	10	10
hold_data	11	11

INFO: [Synth 8-3354] encoded FSM with state register 'state_reg' using encoding 'sequential' in module 'producer'

State	New Encoding	Previous Encoding
idle	00	00
send_ack	01	01
read_data	10	10

INFO: [Synth 8-3354] encoded FSM with state register 'current_state_reg' using encoding 'sequential' in module 'consumer'





It seems that sequential encoding uses less LUTs and FFs, this could be because it uses less bytes and therefore the program doesn't need to generate the elements to simulate them.

CMPE-660 Reconfigurable Computing

Tutorial 2: Handshake and State Machine Tutorial

Student's Name: Diego Diaz

Demo	Point Value	Points Earned	Date
Part 1: Producer and consumer state machine elaborated circuit diagrams. Coding style analysis	25	25	DR 9/11
Part 2: Behavioral simulation and analysis	25	25	
Part 3: Analysis of synthesis reports - number of FFs used for states and registers for two encoding styles	25	25	
Part 4: Post synthesis simulation and	25	25	

Part 4: Post synthesis simulation and analysis for two encoding styles	25	26	↓
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Students will need to demonstrate their understanding of all parts of this Tutorial by answering questions from Instructor/TAs.

Submit this grading sheet on myCourses under Assignemnts → Tut2.